

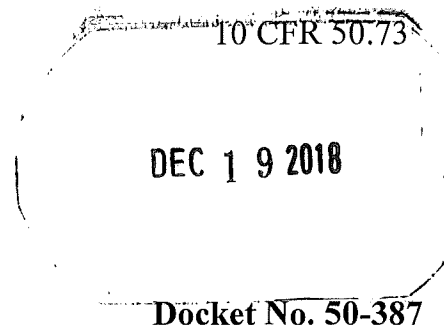
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U. S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387/2018-006-01
UNIT 1 LICENSE NO. NPF-14
PLA-7760**



Attached is a supplement to Licensee Event Report (LER) 50-387/2018-006-00. The LER reported a condition which caused a Standby Liquid Control Pump to fail a surveillance. This condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by technical specifications and 10 CFR 50.73(a)(2)(v)(A) as a condition that could have prevented fulfillment of a safety function.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new regulatory commitments.

Should you have any questions regarding this submittal, please contact Mr. Jason Jennings, Manager – Nuclear Regulatory Affairs at (570) 542-3155.

A handwritten signature in black ink, appearing to be "K. Cimorelli".

K. Cimorelli

Attachment: LER 50-387/2018-006-01

Copy: NRC Region I
Ms. T. E. Hood, NRC Project Manager
Ms. L. H. Micewski, NRC Sr. Resident Inspector
Mr. M. Shields, PA DEP/BRP



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name
Susquehanna Steam Electric Station Unit 1

2. Docket Number
05000387

3. Page
1 OF 3

4. Title
Standby Liquid Control Valve Failed Surveillance Test

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
09	26	2018	2018	006	01	12	19	2018	Facility Name	Docket Number
										05000
									Facility Name	Docket Number
										05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level 095	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☒ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)		

12. Licensee Contact for this LER

Licensee Contact

Nicole Pagliaro, Senior Licensing Specialist – Nuclear Regulatory Affairs

Telephone Number (Include Area Code)

570-542-6578

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☐ No

15. Expected Submission Date

Month Day Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

During the performance of the quarterly flow verification, the 1A Standby Liquid Control (SBLC) Pump could not obtain the Technical Specification required flow of 40 gallons per minute due to the SBLC Injection Pump 'A' Discharge Pressure Relief Valve (PSV148F029A) lifting and being unable to reseal itself. This failed to meet the requirements of Technical Specification Surveillance Requirement 3.1.7.7 and consequently, the 1A SBLC pump was declared inoperable.

This condition is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by technical specifications and 10 CFR 50.73(a)(2)(v)(A) as a condition that could have prevented fulfillment of a safety function. The 'B' train of Standby Liquid Control had been unavailable during the time the 'A' was inoperable.

The cause of the event was determined to be inadequate work instructions for installing the insulation on the associated relief valves. Corrective actions include additional guidance for installing insulation around the relief valves.

There were no actual safety consequences associated with this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station Unit 1	05000-387	YEAR	SEQUENTIAL NUMBER	REV NO.
		2018	006	01

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 95 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 98 percent Rated Thermal Power

As part of the surveillance the 1B Standby Liquid Control Pump was taken out of service. While this did not impact the operability of the 1A Standby Liquid Control Pump it does affect the reporting criteria.

EVENT DESCRIPTION

During performance of quarterly Standby Liquid Control (SBLC) flow verification, on September 26, 2018 at approximately 1023, the 1A SBLC Injection Pump 'A' Discharge Pressure Relief Valve [EIS: RV] lifted after the 1A SBLC pump [EIS:P] was started. The relief valve was unable to reseal itself, resulting in failure to meet Technical Specifications (TS) Surveillance Requirement (SR) 3.1.7.7 minimum required flow rate of 40 gallons per minute (gpm) and consequently, the 1A SBLC pump was declared inoperable.

Maintenance technicians observed that the insulation [EIS: INS] surrounding the relief valve was in contact with its reset arm. Performance data indicates a rapid decrease in SBLC flow approximately 2 minutes after the initial pump start, and is believed to be the point of the relief valve lifting.

The insulation had been re-installed during the Unit 1 Refuel outage on April 27, 2018. This event is being reported as a condition prohibited by technical specification since insulation interference had existed and in the event of relief valve actuation, the 'A' train would have been inoperable longer than the technical specification completion time. Even though both trains were taken out of service as part of a planned surveillance, the condition of the insulation was discovered during the surveillance which led the 'A' train to be considered inoperable. This is being reported as a condition that could have prevented the fulfillment of a safety function since the 'B' SBLC pump was taken out of service during the time frame after the insulation had been re-installed. Specifically, on June 28th for a quarterly surveillance and on July 26 for maintenance.

Maintenance replaced the SBLC Injection Pump 'A' Discharge Pressure Relief Valve and corrected the insulation configuration. As found bench testing was performed on the originally installed relief valve on 10/1/18 which satisfactorily passed lift checks.

CAUSE OF EVENT

The cause of the event was determined to be inadequate work instructions on installing the insulation. There was no information in the routine task instructions advising the insulators on the impact the insulation could have on the moving parts of the valve.



**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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ANALYSIS/SAFETY SIGNIFICANCE

Engineering analysis determined that the Unit 1'B' SBLC pump would have been unaffected by this condition and would have delivered the required flow rate. The 'A' pump discharge check valve 148F033A would have prevented any 'B' loop flow from exiting through the 'A' pressure relief valve, PSV148F029. Since the 'B' loop could have performed the required function, there was no loss of safety function. Note that during quarterly flow surveillance testing both the 'A' and 'B' division are declared inoperable and TS 3.1.7.C is entered. Based on this information, either both SBLC systems were available or both systems were inoperable and the appropriate TS was entered.

This event will not be counted as a safety system functional failure (SSFF) for the NRC performance indicator based on the engineering analysis supporting the system's ability to fulfill the safety function.

CORRECTIVE ACTIONS

Maintenance reconfigured the insulation on the pressure safety valves and shared the lesson learned with the rest of the organization. In addition, Maintenance work instructions were revised to caution insulation installers to ensure the insulation does not impede to travel path of the reset arm.

PREVIOUS OCCURENCES

No previous similar events are known.